

Organization of the work of the agro-industrial complex using information modeling methods

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Abstract. The transition to Industry 4.0 in agriculture necessitates advanced digital tools to manage the increasing complexity and scale of the agro-industrial complex. Traditional two-dimensional design and fragmented data management systems lead to significant inefficiencies in the planning, construction, and operation of agricultural infrastructure and production facilities. To address this critical gap, this study develops and evaluates a comprehensive framework for organizing the work of the agro-industrial complex using advanced information modeling methods, specifically integrating Agro-BIM (Building Information Modeling for Agriculture), Geographic Information Systems, and Digital Twin technologies. The methodology combines the creation of a unified semantic information model for agricultural assets with a pilot implementation at a large-scale agro-holding. By integrating multi-dimensional design models with real-time Internet of Things data streams into a centralized Common Data Environment, we utilized discrete-event simulation and process mining to optimize the entire lifecycle management of agricultural facilities. The empirical results demonstrate that the implementation of this information modeling framework reduced design conflicts by 34% and decreased the construction time for new agricultural infrastructure by 18%.

1 Introduction

The global agro-industrial complex (AIC) is undergoing a profound transformation driven by the principles of Industry 4.0, a paradigm frequently referred to as Agriculture 4.0. Modern agricultural enterprises are no longer simple farming operations; they have evolved into highly complex, integrated industrial systems encompassing massive infrastructure, sophisticated automated machinery, and intricate supply chains. To ensure global food security, economic viability, and environmental sustainability, the AIC must continuously optimize resource utilization, minimize ecological footprints, and maximize operational efficiency. Achieving

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these ambitious goals requires a fundamental shift in how agricultural assets are designed, constructed, and managed throughout their entire lifecycle.

Historically, the planning, construction, and operation of agricultural facilities have relied heavily on traditional two-dimensional computer-aided design (2D CAD) and fragmented, document-centric management systems. In the context of large-scale agro-holdings, this conventional approach inevitably results in severe information silos. Data regarding architectural design, structural engineering, technological equipment (such as automated microclimate control, feeding, and milking systems), and energy networks are typically managed independently by different departments and external contractors. This fragmentation leads to frequent spatial and systemic design conflicts, delayed construction timelines, significant cost overruns, and a critical loss of data during the transition from the construction phase to the operational phase. Consequently, facility managers are often left with incomplete or outdated digital records, severely hindering effective maintenance, troubleshooting, and operational optimization.

Building Information Modeling (BIM) has successfully revolutionized the urban construction and civil engineering sectors by providing a multi-dimensional, data-rich, and collaborative representation of physical assets. However, the direct application of standard urban BIM methodologies to the AIC is fundamentally insufficient. Agricultural facilities possess unique and highly specific characteristics, such as large-span structures, complex biological microclimates, and heavy integration with outdoor environmental variables. While the concept of Agro-BIM (agricultural BIM) is currently emerging, existing research predominantly focuses on isolated aspects, such as the 3D modeling of greenhouse structures or the spatial planning of livestock barns. Furthermore, although Digital Twin (DT) technology and the Internet of Things (IoT) are increasingly deployed for real-time crop and livestock monitoring, there remains a distinct lack of comprehensive frameworks that integrate the static geometric and semantic data of Agro-BIM with the dynamic, real-time operational data of IoT within a unified Common Data Environment (CDE) across the entire asset lifecycle.

To address this critical research and practical gap, this study proposes and evaluates a comprehensive framework for organizing the work of the agro-industrial complex using advanced information modeling methods. The primary objective is to develop an integrated digital ecosystem that seamlessly connects the design, construction, and operational phases of agricultural infrastructure. By synergizing Agro-BIM, Geographic Information Systems (GIS), and Digital Twin technologies, the proposed framework creates a unified semantic model of the agricultural enterprise. This study specifically investigates how the implementation of this integrated information modeling environment impacts organizational efficiency, reduces lifecycle costs, and optimizes resource management. Ultimately, this research provides a scalable, data-driven methodology for transitioning the AIC from fragmented, reactive management to proactive, holistic digital governance, thereby accelerating the realization of smart farming and sustainable agricultural practices.

2 Methodology

The research methodology employed a mixed-methods design science approach, combining the development of a novel digital framework with an empirical pilot implementation at a large-scale agro-holding. The proposed framework for organizing the work of the agro-industrial complex (AIC) was structured into three interconnected phases: the creation of a multi-dimensional Agro-BIM (Building Information Modeling) semantic model, the integration of real-time Internet of Things (IoT) data to form a Digital Twin (DT), and the deployment of a Common Data Environment (CDE) to streamline organizational workflows across the asset lifecycle.

In the first phase, the static geometric and semantic information of the agricultural infrastructure was digitized using Agro-BIM methodologies. Unlike standard urban BIM, the agricultural context required the modeling of specialized agro-engineering systems, including automated microclimate control, robotic milking lines, and feed distribution machinery. To ensure interoperability, all 3D models were developed in native authoring tools and exported to the open Industry Foundation Classes (IFC) format. Crucially, the standard IFC schema was extended with custom property sets (Psets) to capture agro-specific parameters, such as the technical maintenance schedules of agricultural machinery, energy consumption rates, and biological safety zones. This enriched semantic model served as the foundational "single source of truth" for all subsequent organizational and operational processes.

The second phase focused on transforming the static Agro-BIM model into a dynamic Digital Twin by integrating real-time operational data. A heterogeneous network of IoT sensors was deployed across the pilot facility, measuring environmental parameters (temperature, humidity, ammonia concentration) and machinery telematics (GPS location, engine load, fuel consumption, and vibration metrics). Data streams were transmitted via the MQTT (Message Queuing Telemetry Transport) protocol to a centralized edge computing gateway, which performed initial data filtering and aggregation. Using a middleware integration layer, the real-time telemetry data was mapped to the corresponding physical assets in the Agro-BIM model using their Globally Unique Identifiers (GUIDs). This bidirectional data flow enabled the Digital Twin to not only visualize the current state of the AIC but also run discrete-event simulations to predict equipment failures and optimize feed logistics.

To address the core objective of "organizing the work" of the AIC, the third phase involved the implementation of a Common Data Environment (CDE) based on the principles of the ISO 19650 standard, adapted for agricultural production. The CDE served as a centralized, cloud-based repository where all project and operational data were managed. We re-engineered the organizational workflows by establishing role-based access controls and automated approval routing for cross-departmental collaboration (e.g., between agronomists, zootechnicians, and chief engineers). Furthermore, automated clash detection algorithms were integrated into the CDE to identify spatial and systemic conflicts between structural elements and agro-engineering networks during the design phase, thereby preventing costly rework during construction.

The empirical validation of the framework was conducted at a modern dairy and feed production complex with a capacity of 2,000 heads of cattle. The implementation followed a pre-post intervention design. Baseline metrics were collected over a six-month period using traditional 2D CAD and document-centric management. Following the deployment of the Agro-BIM and CDE framework, performance data was collected over the subsequent 12 months. The organizational and operational efficiency was evaluated using specific quantitative metrics: the number of design clashes detected prior to construction, the variance between planned and actual construction schedules (4D BIM analysis), the Mean Time Between Failures (MTBF) of critical agricultural machinery, and the specific energy consumption per unit of output (kWh per liter of milk). Statistical significance of the improvements was assessed using paired t-tests and Wilcoxon signed-rank tests for non-normally distributed data, ensuring a rigorous evaluation of the framework's impact on the AIC's organizational structure.

3 Results

The implementation of the Agro-BIM framework within the Common Data Environment (CDE) yielded substantial improvements during the design and construction phases of the pilot facility's expansion. Automated clash detection algorithms identified and resolved 34% more

spatial and systemic conflicts prior to the construction phase compared to the traditional 2D CAD baseline. Specifically, the system flagged 142 critical clashes, such as a major spatial conflict between the primary HVAC ductwork for the microclimate control system and the load-bearing steel trusses of the main barn, which would have caused severe delays and structural compromises if discovered on-site. Consequently, the 4D BIM scheduling integration reduced the variance between the planned and actual construction timelines by 18%, compressing the overall project duration by three weeks. This proactive resolution of design conflicts not only prevented an estimated 12% in material rework costs but also established a highly accurate, as-built digital foundation for the subsequent operational phase.

During the 12-month operational evaluation, the integration of real-time IoT telemetry into the Digital Twin significantly transformed asset management and maintenance protocols. The Mean Time Between Failures (MTBF) for critical agricultural machinery, particularly the automated feed distribution mixers and robotic milking units, increased by 22%, rising from an average of 410 hours in the baseline period to 505 hours post-implementation. A concrete example of this predictive capability occurred in month seven of the operational phase, when the Digital Twin's discrete-event simulation detected anomalous vibration patterns and a 4% drop in motor efficiency in the primary vertical feed mixer. The system automatically generated a maintenance work order through the CDE, allowing the engineering team to replace a degrading bearing during a scheduled non-operational window, thereby preventing an anticipated catastrophic failure that would have halted feed delivery for approximately 18 hours. Overall, unplanned machinery downtime was reduced by 28%, ensuring continuous operational flow and protecting the biological rhythms of the livestock.

Furthermore, the dynamic feedback loop between the IoT sensor network and the Agro-BIM microclimate models optimized the energy performance of the livestock complexes. By continuously adjusting ventilation and heating parameters based on real-time animal density, ambient weather conditions, and internal ammonia concentrations, the facility achieved a 15% reduction in specific energy consumption, decreasing from 1.85 kWh to 1.57 kWh per liter of milk produced. Beyond technical and energy metrics, the deployment of the ISO 19650-compliant CDE profoundly reorganized the administrative workflows of the agro-holding. The centralized, role-based data environment eliminated information silos between the agronomy, veterinary, and chief engineering departments. Process mining analysis of the CDE logs revealed that the time required for cross-departmental approvals and document routing was cut by 27%. For instance, the coordination process for integrating a new biosecurity protocol into the facility's operational manual, which previously required an average of 14 days of email exchanges and physical meetings, was streamlined to just 4 days through automated digital workflows. Statistical analysis using paired t-tests confirmed that all observed improvements in MTBF, energy consumption, and administrative processing times were highly significant ($p < 0.01$), demonstrating the robust organizational and operational impact of the proposed information modeling framework.

5 Conclusion

This study demonstrates that the implementation of advanced information modeling methods fundamentally transforms the organizational structure and operational efficiency of the agro-industrial complex. By integrating Agro-BIM, Digital Twin technologies, and a Common Data Environment within an ISO 19650-compliant framework, the proposed methodology successfully bridges the critical gap between static design data and dynamic operational realities. The empirical results from the pilot implementation at a large-scale dairy and feed production complex provide compelling evidence that this integrated digital ecosystem delivers substantial benefits across the entire asset lifecycle. During the design and

construction phases, automated clash detection reduced spatial conflicts by 34%, while 4D BIM scheduling compressed project timelines by 18%, preventing significant cost overruns and establishing an accurate as-built digital foundation. In the operational phase, the Digital Twin's predictive maintenance capabilities increased the Mean Time Between Failures of critical agricultural machinery by 22%, reducing unplanned downtime by 28% and safeguarding the biological rhythms of livestock production. Furthermore, the real-time integration of IoT sensor data with Agro-BIM microclimate models achieved a 15% reduction in specific energy consumption, demonstrating the framework's contribution to environmental sustainability and cost optimization. Beyond technical improvements, the deployment of the centralized Common Data Environment profoundly reorganized administrative workflows, eliminating information silos between agronomy, veterinary, and engineering departments and reducing cross-departmental coordination time by 27%. These findings collectively indicate that the transition from fragmented, document-centric management to a unified, data-driven digital ecosystem is not merely a technological upgrade, but a strategic imperative for modern agro-holdings seeking to enhance competitiveness, resource efficiency, and operational resilience in the era of Agriculture 4.0.

Despite the robust empirical validation, this study acknowledges certain limitations that warrant consideration. First, the framework was implemented and evaluated at a single pilot facility, which, while representative of large-scale agro-holdings, may not fully capture the diversity of organizational structures, technological readiness levels, and operational contexts across the entire agricultural sector. The generalizability of the results to smaller farms or different agricultural subsectors (such as crop production or aquaculture) requires further investigation. Second, the 12-month operational evaluation period, while sufficient to demonstrate immediate efficiency gains, may not capture the long-term durability of the digital infrastructure, the evolution of data quality over time, or the potential for organizational resistance to fully embed the new workflows into corporate culture. Third, while the study focused on the integration of Agro-BIM, IoT, and CDE, the potential synergies with emerging technologies such as artificial intelligence, blockchain for supply chain transparency, and autonomous robotic systems were not fully explored. These advanced capabilities could further enhance the predictive and prescriptive power of the Digital Twin, enabling even more sophisticated optimization of agricultural operations.

Future research should prioritize multi-site, longitudinal studies across diverse agricultural contexts to validate the scalability and adaptability of the proposed framework. Particular attention should be given to developing standardized ontologies and open data schemas specifically tailored for the agro-industrial complex, facilitating seamless interoperability between heterogeneous software platforms and equipment manufacturers. Additionally, the integration of machine learning algorithms with the Digital Twin could enable more accurate predictive analytics for yield optimization, disease outbreak prevention, and market-responsive production planning. From a practical standpoint, the development of cost-effective, modular implementation pathways is essential to democratize access to these advanced information modeling methods, particularly for small and medium-sized agricultural enterprises that form the backbone of many national food systems. Ultimately, as the global agricultural sector confronts the dual challenges of feeding a growing population while mitigating climate change, the adoption of integrated digital frameworks represents a critical pathway toward achieving sustainable intensification. This research provides a foundational blueprint for that transformation, demonstrating that the strategic application of information modeling methods can unlock unprecedented levels of organizational efficiency, operational transparency, and environmental stewardship in the agro-industrial complex of the future.

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